Exercise 2.3. Prove the following. For each, n is an integer.

(a) If n is odd, then $n^2 + 4n + 9$ is even.

(b) If n is odd, then n^3 is odd.

(c) If n is even, then $n + 1$ is odd.

(d) 1 divides

Solution:

For this exercise, we again use:

- If n is **even**, then $n = 2k$ for some $k \in \mathbb{Z}$.
- If n is **odd**, then $n = 2k + 1$ for some $k \in \mathbb{Z}$.

(a) If n is odd, then $n^2 + 4n + 9$ is even

Proof

Assume n is odd. Then

$$n = 2k + 1$$

for some integer k.

Substitute this into the expression:

$$n^2 + 4n + 9$$

$$= (2k + 1)^2 + 4(2k + 1) + 9$$

Expand:

$$= 4k^2 + 4k + 1 + 8k + 4 + 9$$

$$= 4k^2 + 12k + 14$$

Factor out 2:

$$= 2(2k^2 + 6k + 7)$$

Since $2k^2 + 6k + 7$ is an integer, the expression has the form $2m$.

Therefore,

$$\boxed{n^2 + 4n + 9 \text{ is even.}}$$

(b) If n is odd, then n^3 is odd

Proof

Assume n is odd. Then

$$n = 2k + 1$$

for some integer k.

Therefore,

$$n^3 = (2k+1)^3$$

Expanding:

$$= (2k+1)(2k+1)(2k+1)$$

or directly,

$$= 8k^3 + 12k^2 + 6k + 1$$

Factor out 2 from all terms except 1:

$$= 2(4k^3 + 6k^2 + 3k) + 1$$

Since

$$4k^3 + 6k^2 + 3k$$

is an integer, n^3 has the form

$$2m + 1.$$

Therefore,

$$\boxed{n^3 \text{ is odd.}}$$

(c) If n is even, then $n + 1$ is odd

Proof

Assume n is even. Then

$$n = 2k$$

for some integer k.

Thus,

$$n + 1 = 2k + 1$$

Since k is an integer, this has the form $2m + 1$, which is the definition of an odd integer.

Therefore,

$$\boxed{n + 1 \text{ is odd.}}$$